

A Note on the Structure of Locally Outerplanar Graphs

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Abstract

A convex geometric graph is called *locally outerplanar* if there is no path of length three intersects itself. In [D. Boutin, *Convex Geometric Graphs with No Short Self-intersecting Path*, *Congressus Numerantium* 160 (2003) 205–214.] and [D. Boutin, *Structure and Properties of Locally Outerplanar Graphs*, *Journal of Combinatorial Mathematics and Combinatorial Computing* 60 (2007) 169–180.], Boutin states the results of the degeneracy, a bound on the number of edges, and a structural property on locally outerplanar graphs based on the concept of “minimal corner pairs”. In this paper, we show that “minimal corner pair” is not well-defined and give a counterexample to disprove the structural property. Revised simpler proofs for the degeneracy and a bound on the number of edges are also proposed.

Keywords: geometric graphs; locally outerplanar graphs; self-intersecting paths

1 Introduction

A graph $G = (V, E)$ drawn in the plan in which the vertex set V consists of points in general position and the edge set E consists of straight-line segments between points of V is called a *geometric graph*. A common question is :“What is the maximum number of edges that a geometric graph on n vertices can have without containing a forbidden geometric subgraph?”. There are some forbidden subgraphs that have been well-studied. For example, sets of pairwise crossing edges [4,11], sets of pairwise disjoint edges [1, 5], self-intersecting paths [2, 3, 8], noncrossing paths [7, 10], and self-intersecting cycles [9]. Readers can refer to [6] for a detailed survey.

In [2], Boutin defined *locally outerplanar graphs* as the geometric graphs with vertices located on convex positions and have no *self-intersecting paths* of length three. A self-intersecting path is a

path in a geometric graph, in which some edges intersects. In [2,3], Boutin states the following results:

- A structural property: a convex geometric graph is locally outerplanar if and only if it can be built by layering vertex disjoint outerplanar graph drawings and tying these together with nonadjacent vertices from additional outerplanar graph drawings.
- The degeneracy: every locally outerplanar graph contains a vertex of degree less than or equal to two.
- A bound on the number of edges: a locally outerplanar graph with at least one intersection contains at most $2n - 6$ edges, where n is the number of vertices.

Boutin adopt the notion of *minimal corner pair* to prove the above statements. However, the term “minimal corner pair” is not well-defined, which results in the incorrectnesses of the proofs in [2,3]. We shall show in Section 3 that the structural property does not hold via a counterexample. The results of the degeneracy and the bound on the number of edges are correct. We propose proofs based on only the fundamental definition of locally outerplanar graphs.

The paper is organized as follows. In Section 2, we define the notation and the terminology. In Section 3, we formally state the structural property in [3] and gives a counterexample to disprove it. The proofs for the degeneracy and the bound on the number of edges are given in Section 4. Finally, concluding remarks are given in Section 5.

2 Preliminaries

In this paper, the terminology is consistent with that used in [3]. Given a graph G , we denote the vertex set and the edge set of G by $V(G)$ and $E(G)$, respectively. For any graph G , the corresponding convex geometric drawing is denoted by $\bar{G}$. Let n be the number of vertices in G . Conventionally, in $\bar{G}$, the vertices are labeled with

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$1, 2, \dots, n$ in a clockwise direction and can be referred to as their labels. An edge with end vertices u and v is denoted by $e_{u,v}$. The interval $[a, b]$ denotes the set of vertices lie clockwise between a and b , including both a and b . The subgraph induced by $[a, b]$ is denoted by $G[a, b]$. The *cyclic distance* $\hat{d}(u, v)$ between vertices u and v is defined as $\hat{d}(u, v) \equiv v - u \pmod{n}$.

A path $P = (a_1, a_2, \dots, a_k)$ is called a *simple self-intersecting path* if e_{a_1, a_k} and $e_{a_2, a_{k-1}}$ is the only intersection caused by the edges of P and $P' = (a_2, a_3, \dots, a_{k-1})$ does not cross itself. The pair (a_2, a_{k-1}) is called a *corner pair* and is denoted by $\langle a_2, a_{k-1} \rangle$. For a corner pair $\langle a, b \rangle$, the subgraph $\bar{H}_{a,b}$ denotes a connected component containing a and b of the subgraph induced by $[a, b]$. A *minimal corner pair* is a corner pair $\langle a, b \rangle$, which is the only corner pair within $\bar{H}_{a,b}$. The *degree* of a vertex v is the number of edges incident with v and is denoted by $\deg(v)$. The *minimum degree* of a graph G is defined as $\delta(G) = \min_{v \in V(G)} \deg(v)$. The *neighborhood* of a vertex v is defined as $N(v) = \{u : u \in V(G) \text{ and } e_{uv} \in E\}$. For each $u \in N(v)$, u is called a *neighbor* of v .

Property 1. [3] Any subgraph drawing of a locally outerplanar graph is locally outerplanar.

3 A disproof of the structural property

In [3], Boutin states that “A convex geometric graph is locally outerplanar if and only if it can be built by layering vertex disjoint outerplanar graph drawings and tying these together with nonadjacent vertices from additional outerplanar graph drawings.”, and gave a proof by using the concept of minimal corner pairs. Due to the inappropriate definition of a minimal corner pair, both the proof and the theorem are incorrect. The incorrectness is verified via a counterexample. A formal statement of the result in [3] is given in Theorem 1.

Theorem 1 (See [3]). *A geometric graph $\bar{G}$ is locally outerplanar if and only if it contains two sequences of induced convex geometric subgraphs $\bar{H}_1, \dots, \bar{H}_s$ and $\bar{G} = \bar{G}_0, \bar{G}_1, \dots, \bar{G}_s$ so that $\bar{G}_s$ is a union of vertex disjoint outerplanar graph drawings, and for each $i \geq 1$*

1. $\bar{H}_i$ and $\bar{G}_i$ provide an edge partition of $\bar{G}_{i-1}$
2. $\bar{H}_i$ is an outerplanar graph drawing
3. $V(\bar{G}_i) \cap V(\bar{H}_i) = \{u_i, v_i\}$ where

- (a) $V(\bar{H}_i) \subseteq [u_i, v_i]$ as a subset of $V(G)$
- (b) u_i and v_i are consecutive vertices on the convex hull of $\bar{H}_i$ and the shortest path between them has length at least 2.
- (c) in $\bar{G}_i$, $[u_i, v_i]$ does not contain the terminal vertex of any edge with initial vertex in the first neighborhood of u_i or v_i .

The graph $\bar{G}$ in Figure 1 violates the necessity of Theorem 1. It is noted that $\bar{G}$ contains no minimal corner pair, even though $\bar{G}$ is connected and has intersections.

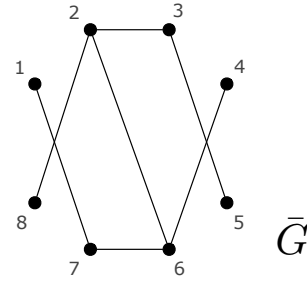


Fig. 1. A locally outerplanar has two corner pairs $\langle 2, 7 \rangle$ and $\langle 6, 3 \rangle$, but no minimal corner pair.

Without loss of generality, we assume that $\bar{H}_i$ contains at least one edge. In the following, we show that $\bar{G}_1$ and $\bar{H}_1$ do not exist, which disprove the necessity of Theorem 1. By condition 3 in Theorem 1, $V(\bar{G}_1) \cap V(\bar{H}_1) = \{u_1, v_1\}$. Thus, there are 56 possible cases for the ordered pair (u_1, v_1) . Because of symmetry, we consider only the 28 pairs with $u_1 \in \{1, 2, 3, 4\}$. Moreover, by condition 3(b), u_1 and v_1 are not adjacent. It follows that $(u_1, v_1) \notin \{(1, 7), (2, 3), (2, 6), (2, 8), (3, 5), (4, 6), (6, 7)\}$. Furthermore, $\hat{d}(u_1, v_1) \neq 1$ since otherwise, either that there is no edge between $[u_1, v_1]$ or that u_1 is adjacent to v_1 . As a result, it remains to consider the 15 possible pairs $\{(1, 3), (1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 7), (3, 1), (3, 6), (3, 7), (3, 8), (4, 1), (4, 2), (4, 6), (4, 7), (4, 8)\}$ of (u_1, v_1) .

According to condition 3 and 3(c), all vertices which are reachable from u_1 or v_1 in $G[u_1, v_1]$ are contained in $\bar{H}_1$. Based on this observation, we have

- For $(u_1, v_1) \in \{(1, 3), (1, 4), (2, 4)\}$, we have $e_{2,3} \in E(\bar{H}_1)$ whose inclusion violates condition 3.

- For $(u_1, v_1) \in \{(1, 5), (2, 5)\}$, we have $e_{2,3} \in E(H_1)$ and $e_{2,6} \in E(G_1)$, which violates condition 3.
- For $(u_1, v_1) \in \{(1, 6), (2, 7), (3, 1), (3, 6), (3, 7)\}$, we have $\{e_{3,5}, e_{4,6}\} \subseteq E(H_1)$, which violates condition 2.
- For $(u_1, v_1) \in \{(4, 1), (4, 7), (4, 8)\}$, we have $e_{4,6} \in E(H_1)$ and $e_{2,6} \in E(G_1)$, which violates condition 3.
- For $(u_1, v_1) = (3, 8)$, we have $e_{2,6} \in E(G_1)$, which violates condition 3(c).
- For $(u_1, v_1) = (4, 2)$, we have $\{e_{1,7}, e_{2,8}\} \subseteq E(H_1)$, which violates condition 2.

Due to the above analysis, it follows that the necessity of Theorem 1 in [3] is incorrect.

4 The degeneracy and a bound on the number of edges

Since there may be no minimal corner pair in a connected locally outerplanar graph, the proofs provided in [2] are incorrect. In the following, we prove that a locally outerplanar graph is 2-degenerate in Theorem 2. Based on Theorem 2, a modified proof of the bound on the number of edges in [2] is given in Theorem 3.

Theorem 2. *For a locally outerplanar graph $\bar{G}$, we have $\delta(G) \leq 2$.*

Proof. Suppose to the contrary that $\delta(G) \geq 3$. There is a vertex pair (u', v') satisfying

- $e_{u',v'} \in E$;
- $\exists w' \in N(u') \cup N(v')$ such that $N(w') \subseteq [u', v']$.

Such a pair is called a *hat*. It is noted that “hat” is well-defined since (x, x_2) satisfies both restrictions, where $x \in V$ and x_2 is the second neighbor of x . Let $H = \{(u', v') : (u', v') \text{ is a hat}\}$ and

$$(u, v) = \arg \min_{(u', v') \in H} \hat{d}(u', v').$$

Because of symmetry, let w be a vertex satisfying $w \in N(u)$ and $N(w) \subseteq [u, v]$. Since $\delta(G) \geq 3$, there are distinct vertices w_1 and w_2 which belong to $N(w) - \{u, v\}$. There are three possible cases: (i) $w_1, w_2 \in [u, w]$; (ii) $w_1, w_2 \in [w, v]$; (iii) $w_1 \in [u, w]$ and $w_2 \in [w, v]$. For case (i), since $\bar{G}$ is locally outerplanar, we have $N(w_1) \subseteq [u, w]$. Moreover, $uw \in E$, $w_1 \in N(w)$, and $N(w_1) \subseteq [u, w]$ imply that (u, w) is a smaller hat, which contradicts the minimality. For cases (ii) and (iii), there are smaller hats (w, v) and (u, w) , respectively. The reasons are similar to that in (i). $\square$

Theorem 3. *Let $\bar{G}$ be a locally outerplanar graph with n vertices. If $\bar{G}$ contains at least one intersection, then $e(\bar{G}) \leq 2n - 6$.*

Proof. We prove this theorem by induction on n . Since $\bar{G}$ contains at least one intersection, we have $n \geq 4$.

For $n = 4$, there are exactly two edges that form an intersection. Thus, $e(\bar{G}) = 2 = 2 \cdot 4 - 6$. Suppose that the statement holds for $n = k$. For $n = k + 1$, consider the subgraph $\bar{G} - x$, where x is a vertex with $\deg(x) = \delta(\bar{G})$. By Property 1, $\bar{G} - x$ is still locally outerplanar, and therefore $e(\bar{G} - x) \leq 2(n - 1) - 6$. Together with Theorem 2, we have

$$\begin{aligned} e(\bar{G}) &= e(\bar{G} - x) + \deg(x) \\ &\leq 2(n - 1) - 6 + \deg(x) \\ &\leq 2(n - 1) - 6 + 2 \\ &= 2n - 6, \end{aligned}$$

which completes the proof. $\square$

5 Concluding remarks

Boutin states the results of the degeneracy, a bound on the number of edges, and a structural property on locally outerplanar graphs based on the concept of “minimal corner pairs”. However, the term “minimal corner pair” is not well-defined, which results in the incorrectnesses of the proofs in [2, 3]. Furthermore, Boutin derives a tighter upper bound on the number of edges of a locally outerplanar graph (Corollary 1.2. [3]) based on Theorem 1. However, due to the incorrectness, the bound is incorrect. We would like to propose a necessary and sufficient condition for locally outerplanar graphs, and, based on the condition, derive a tighter bound on the number of edges.

Acknowledgement

Chun-Yu Tseng and Hung-Lung Wang were supported in part by NSC grants 99-2115-M-141-003-MY2 from the National Science Council, Taiwan.

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